A First Course in Causal Inference Ch. 9

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Ch. 9 Bridging Finite and Super Population Causal Inference

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Definition 9.1 internal validity

The statistical analysis is valid for the study samples at hand.

Definition 9.2 external validity

The statistical analysis is valid for a broader population beyond the study samples.

- Do finite population results generalize to a bigger population?
- The external validity of the results depends on the sampling process of the units
  - If the finite population represents a sample of a larger population, the results also have external validity.
- The results based on randomization inference may not generalize; it ensures only internal validity.

Ch. 9 Bridging Finite and Super Population

To rigorously discuss this issue, we introduce a frmework with two levels of randomness:

- Randomness due to the sampling process of the units;
- Randomness due to the random assignment of the treatment.

Actually, this framework does not really solve the problem mentioned above, but it is beneficial for the following two purposes:

- it gives a different perspective on randomized experiments;
- it serves as a bridge between experimental studies and observational studies.

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